

Thus, embedded systems are generally systems within systems, hence they generally cannot function on their own. Consider a Digital Set Top Box (DST) which might be sitting in your home. The A/V decoder i.e. the digital audio/video decoding system is an embedded system and is a very important part of the DST. Taking a single multimedia stream as input, the A/V decoder produces sound and video as output. But these signals received from the satellite by the DST contain multiple streams, which the A/V decoder doesn't like, and so comes another embedded system to save the day – the transport stream decoder. This decoder takes the incoming multimedia streams and splits it into separate channels and gives only the required channel to the A/V decoder. And so you have your fully functioning DST, all thanks to embedded systems.

Now we did say that embedded systems generally cannot or don't function on their own, but there are exceptions. These are stand-alone embedded systems. They take input, process them, and produce the desired output. The input here can be commands from a person, such as pressing of a switch, or they could be electrical signals from transducers. This output will then be used to drive another system such as the display on a microwave oven. A network router is a stand-alone embedded system. Built using a specialised communications processor, memory, a number of network access, and network ports, the network router routes packets from one port to another based on a programming algorithm, and hence is a stand-alone embedded system.

So can a PC that is only used for browsing be called an embedded system? Well, to be able to understand this we need to be able to truly understand what an embedded system is considering its characteristics and various perspectives, so we'll get to that in a minute. First, let us see how a computer is different from an embedded system.

First, what is a computer?

Now, we know that you may know this already, but bear with us, there is a point to this. A computer is a system that will have the following or more components:

- A microprocessor
- A large memory comprising the following two types:
 - Primary Memory (or semiconductor memories – RAM, ROM, and fast accessible caches)